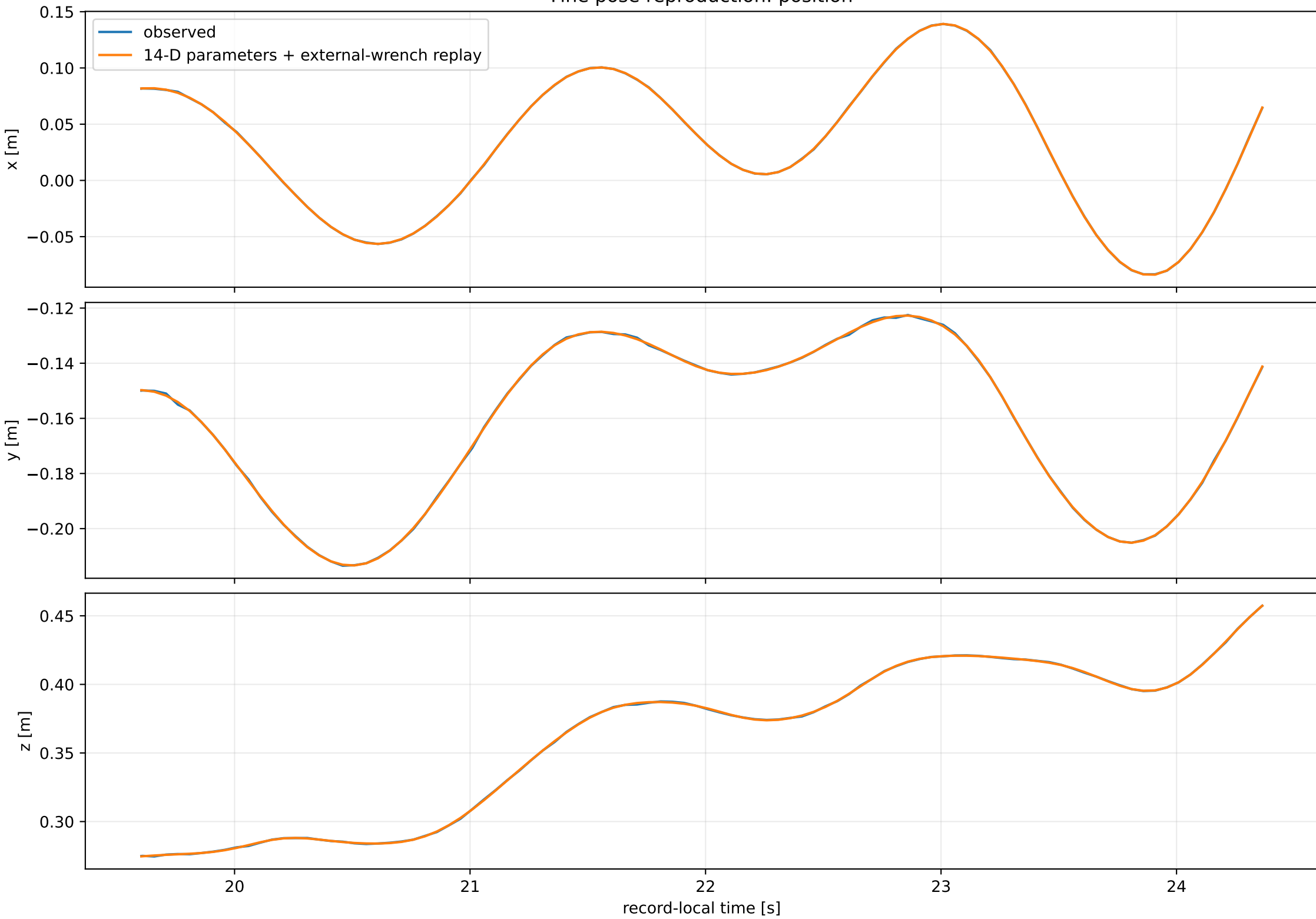
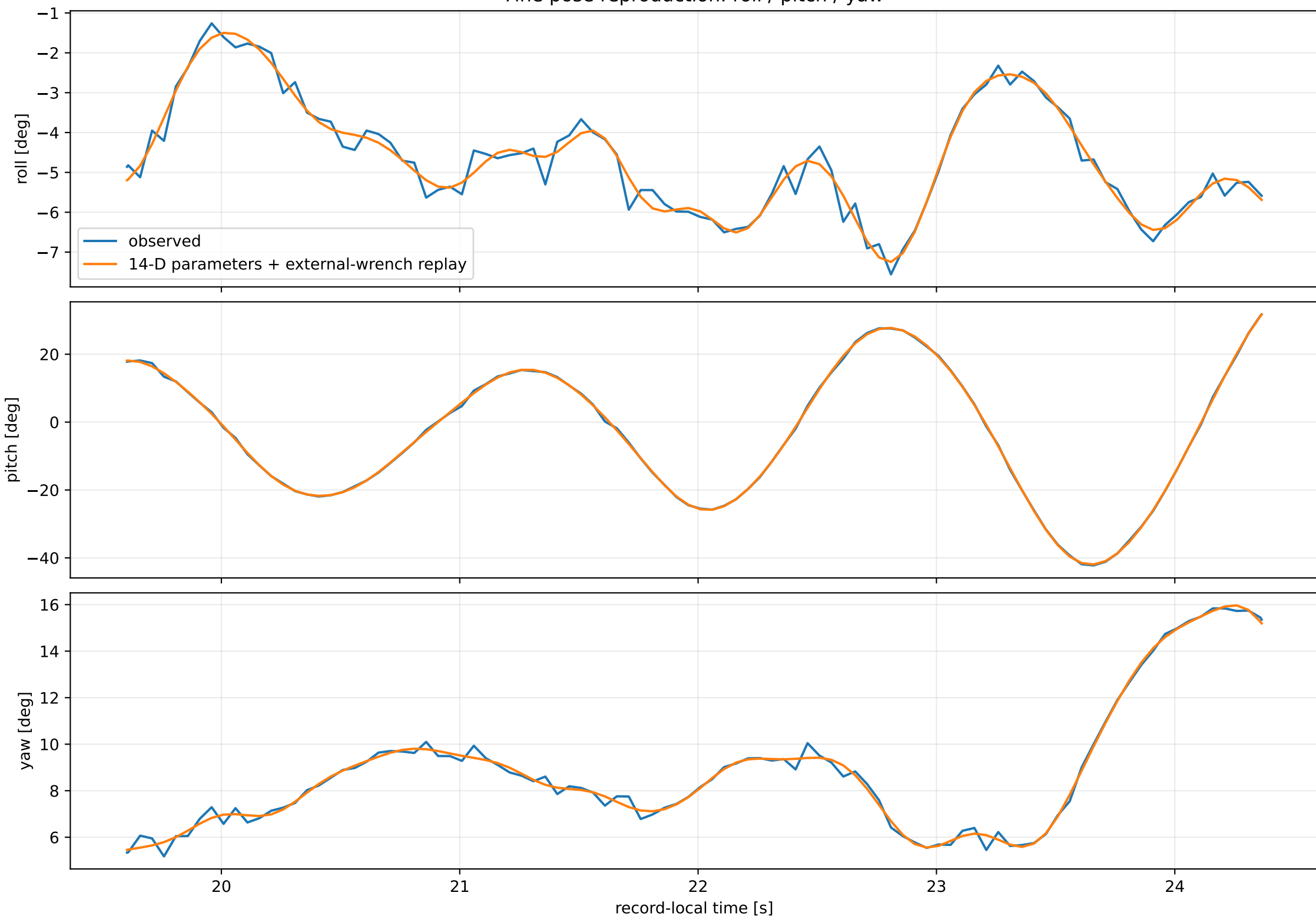


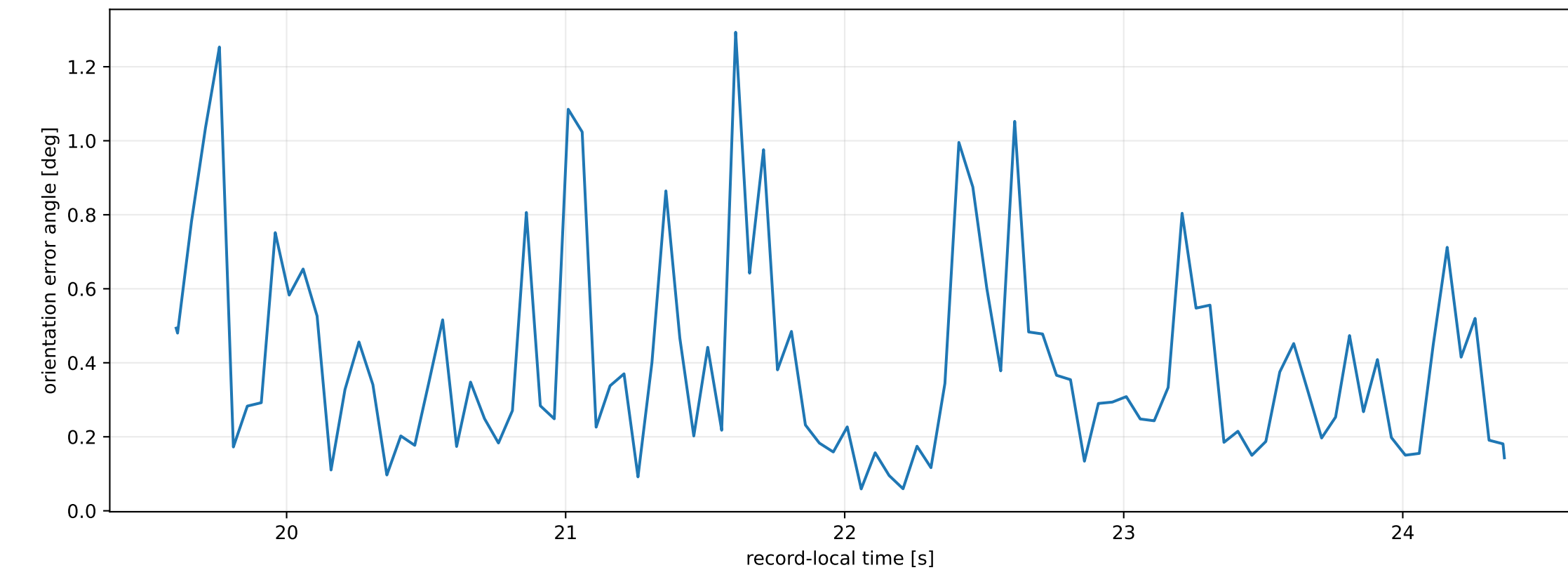
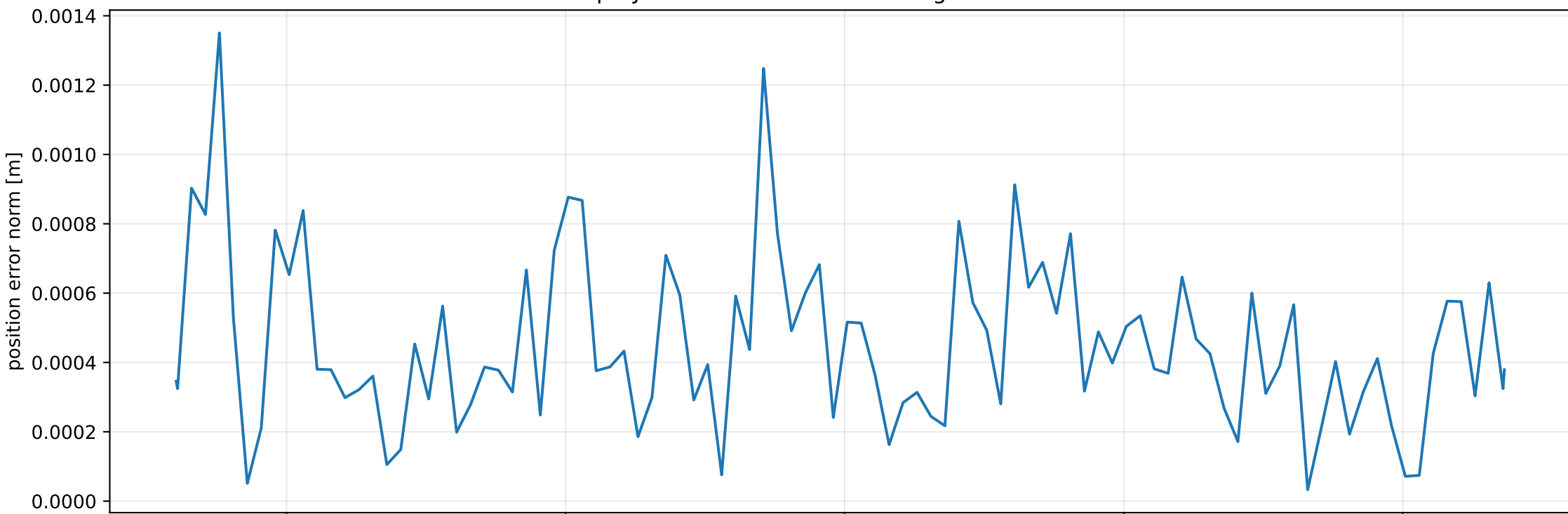
Fine pose reproduction: position



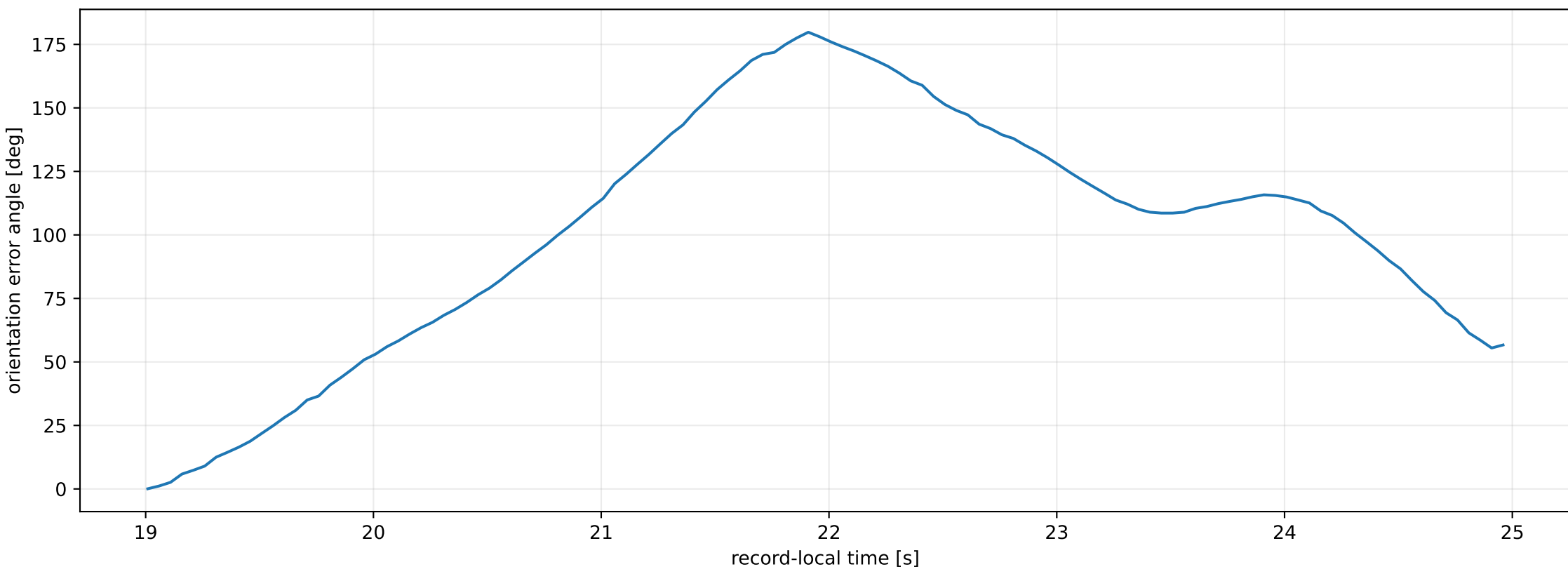
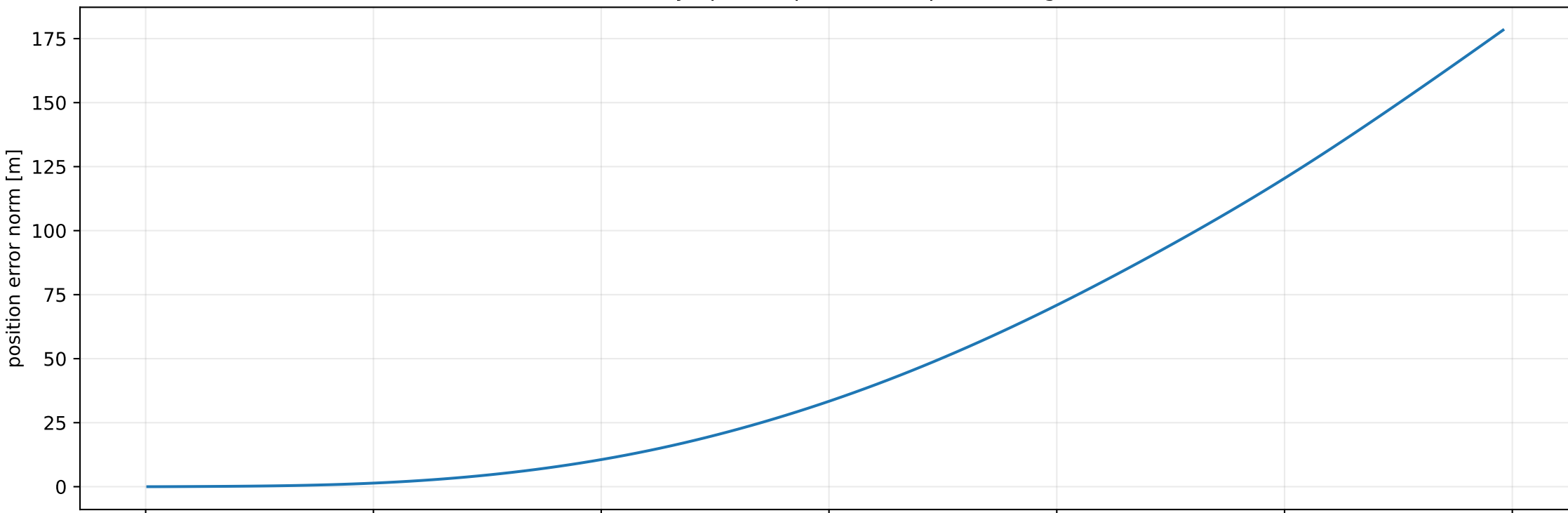
Fine pose reproduction: roll / pitch / yaw



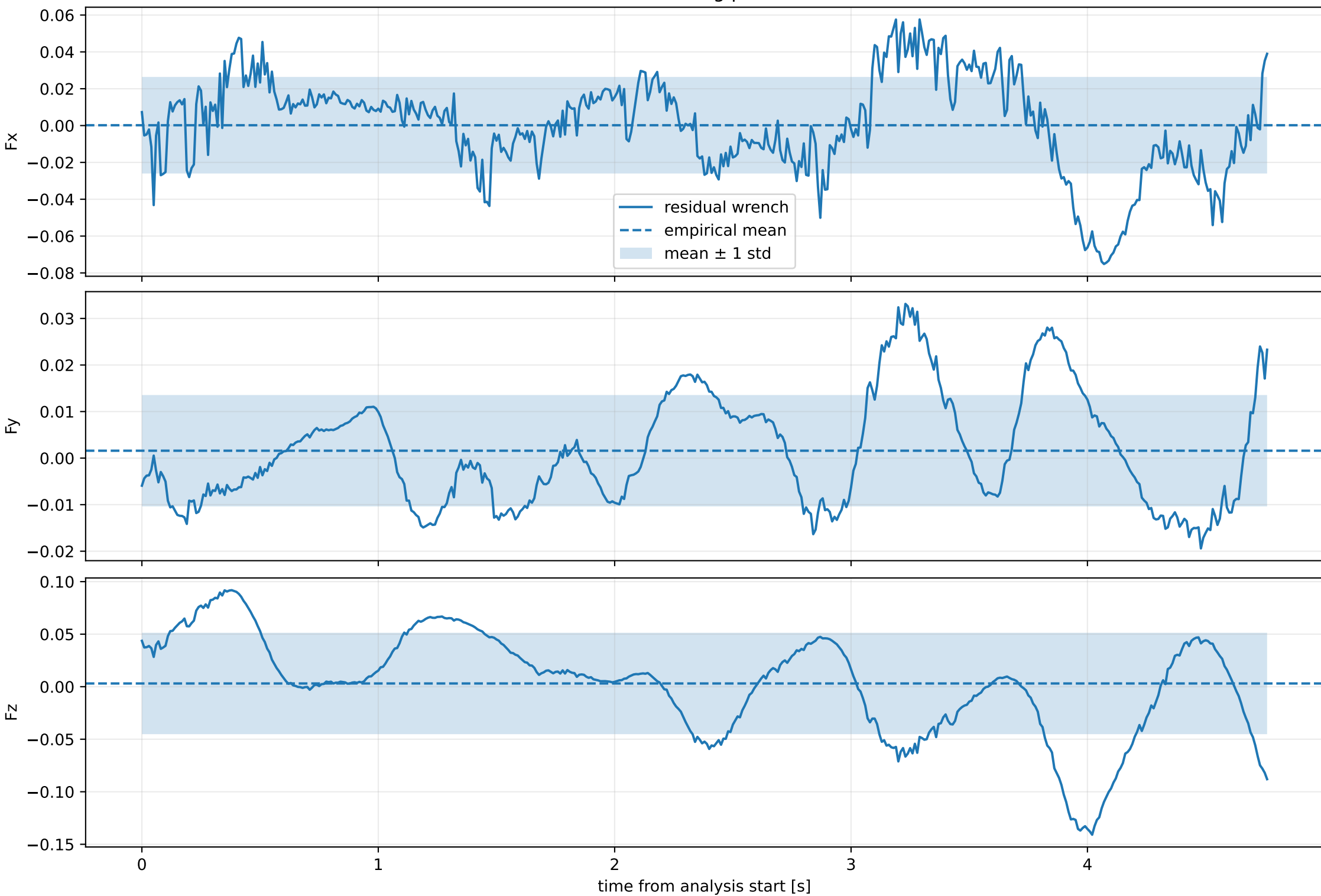
Fine replay error after the 14-D change — lower is better



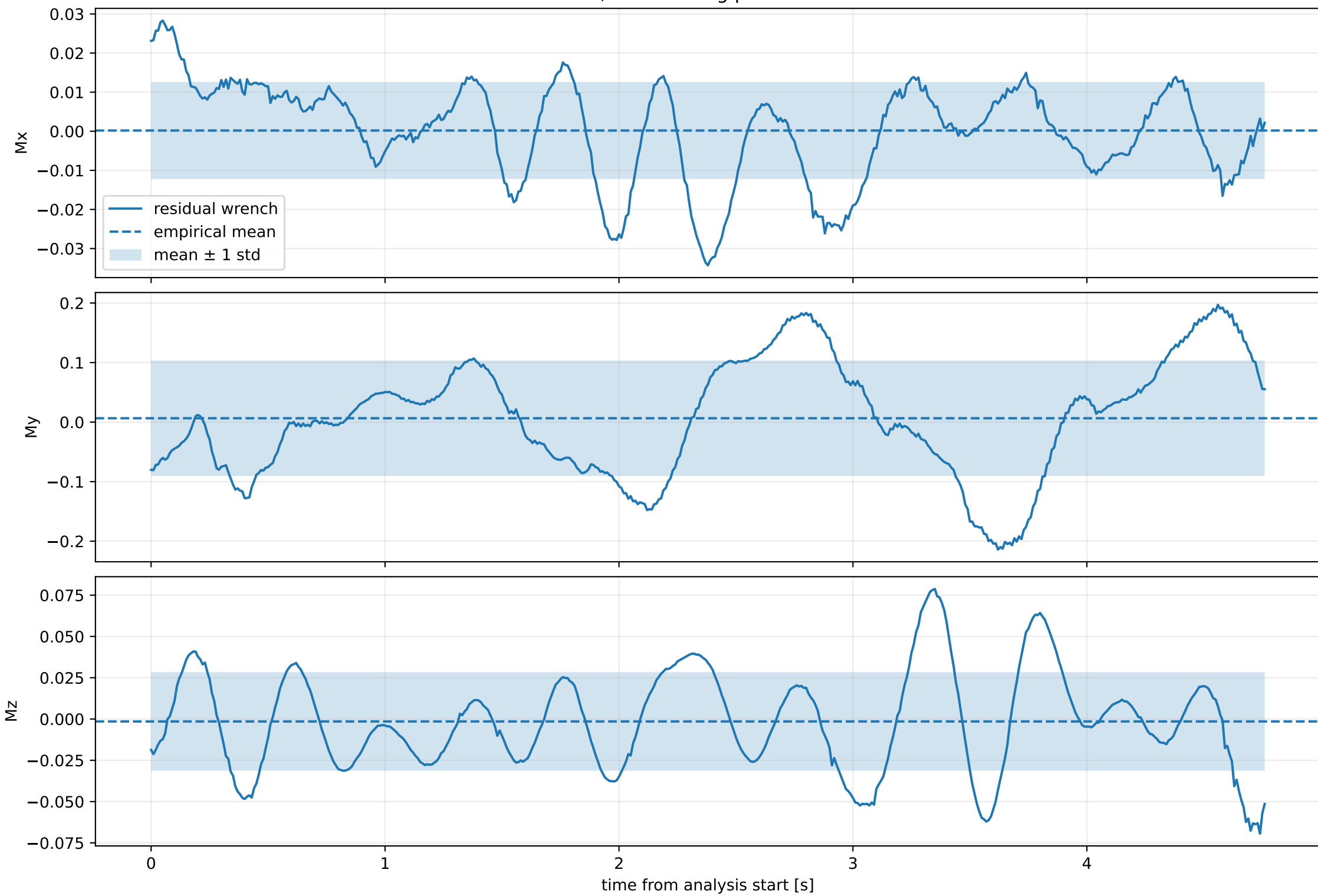
Parameter-only open-loop rollout (separate diagnostic)



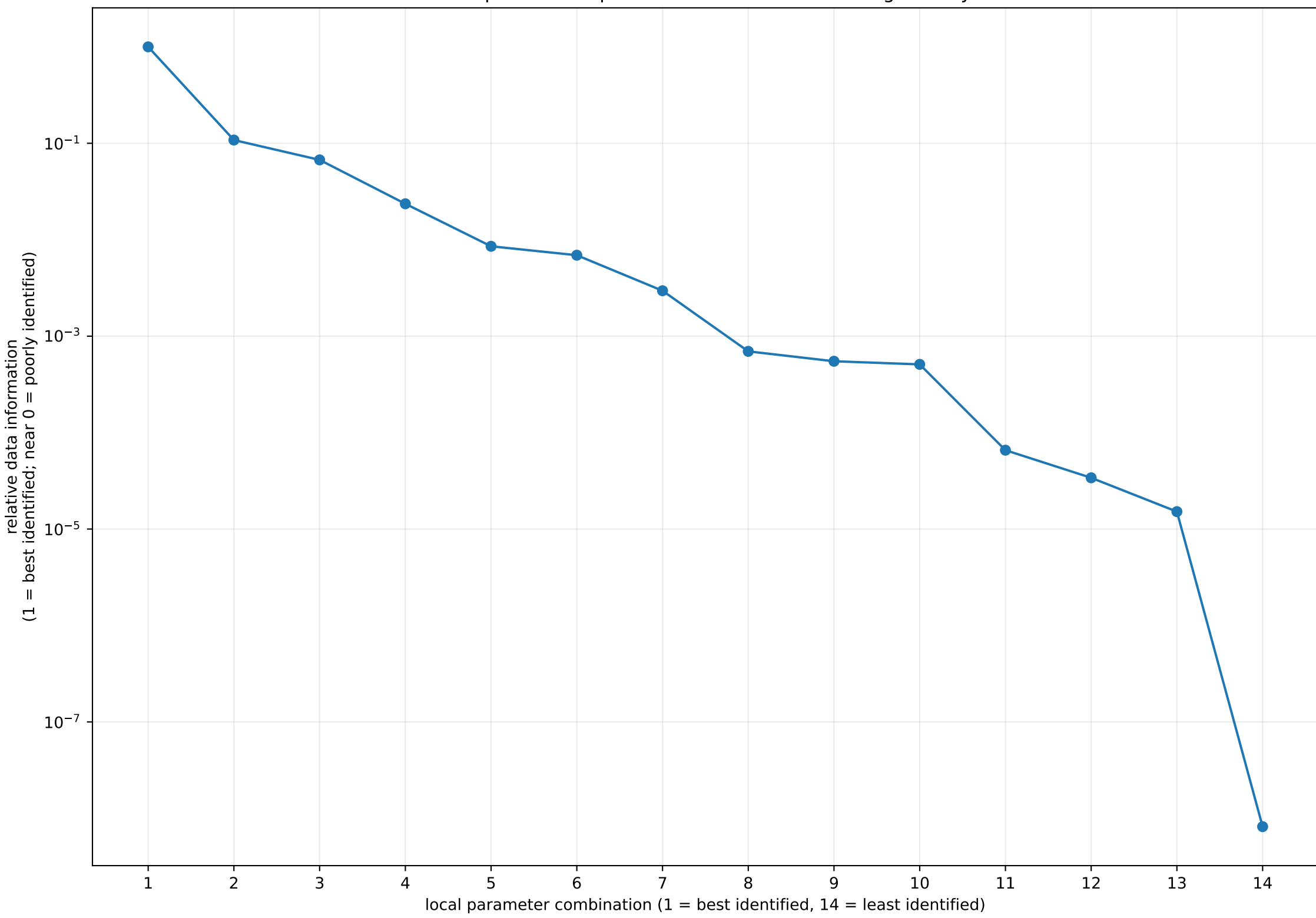
Dimensionless residual force: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



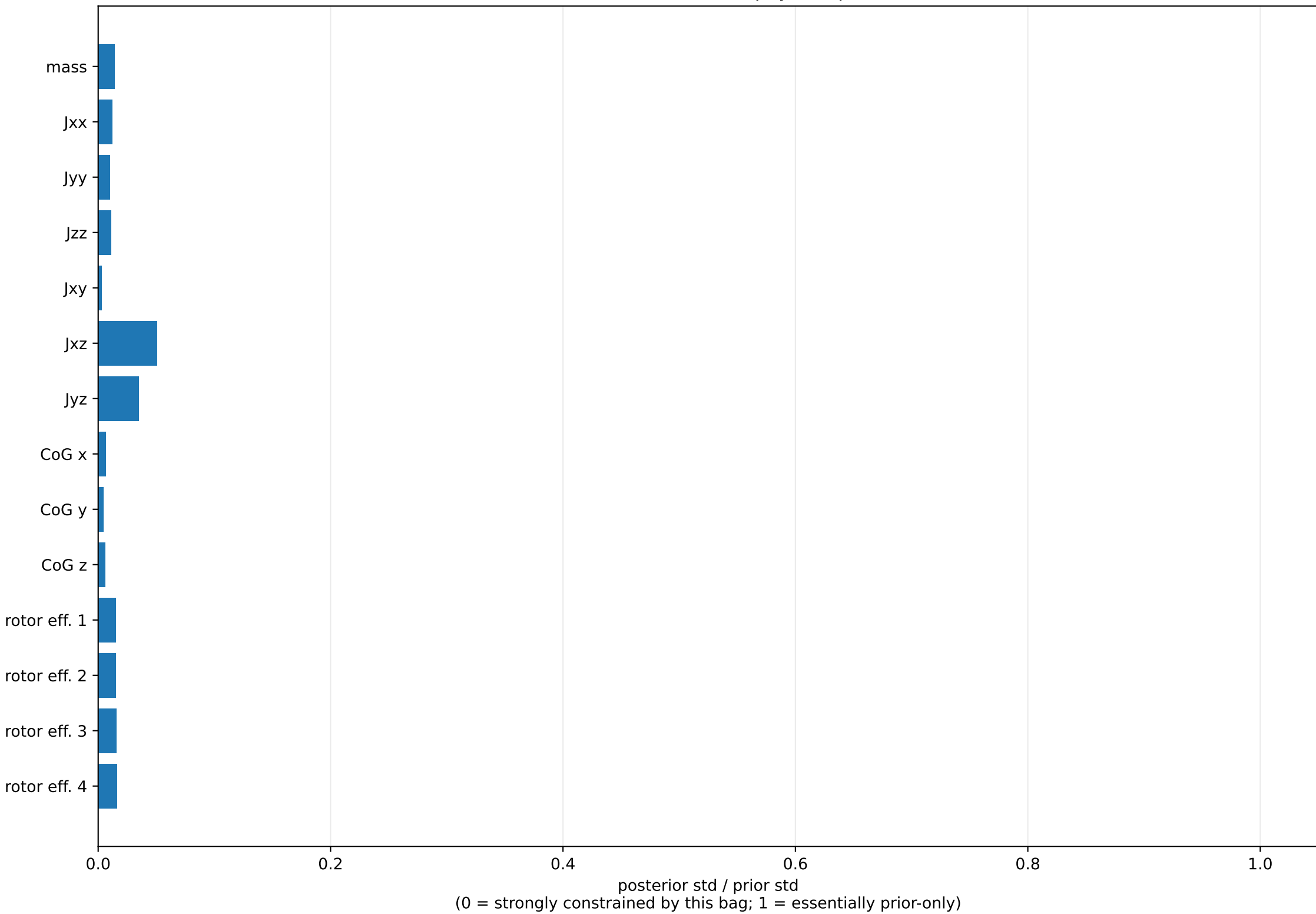
Dimensionless residual torque: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



What parameter-space combinations did this bag identify?

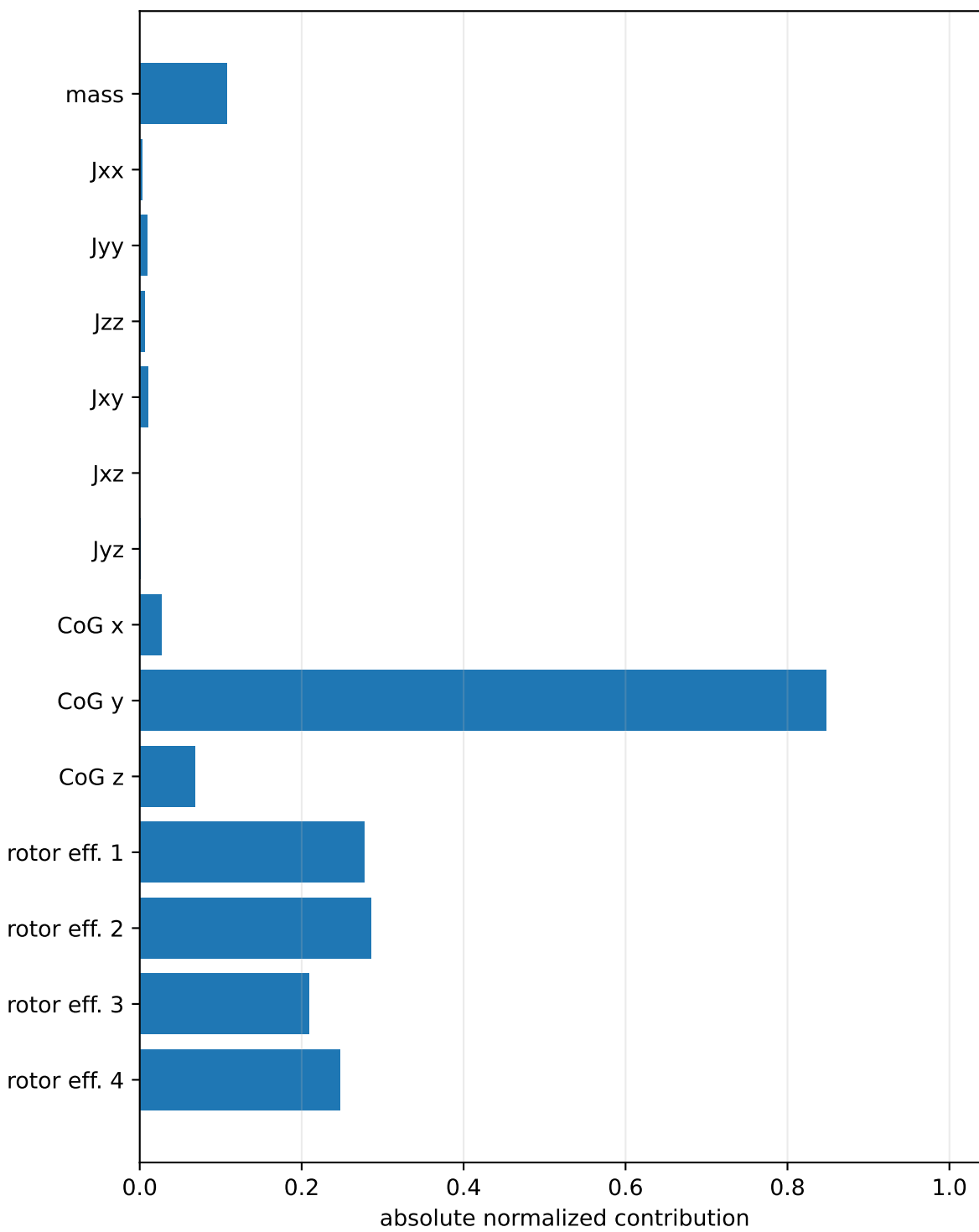


How confident are we in each physical parameter?

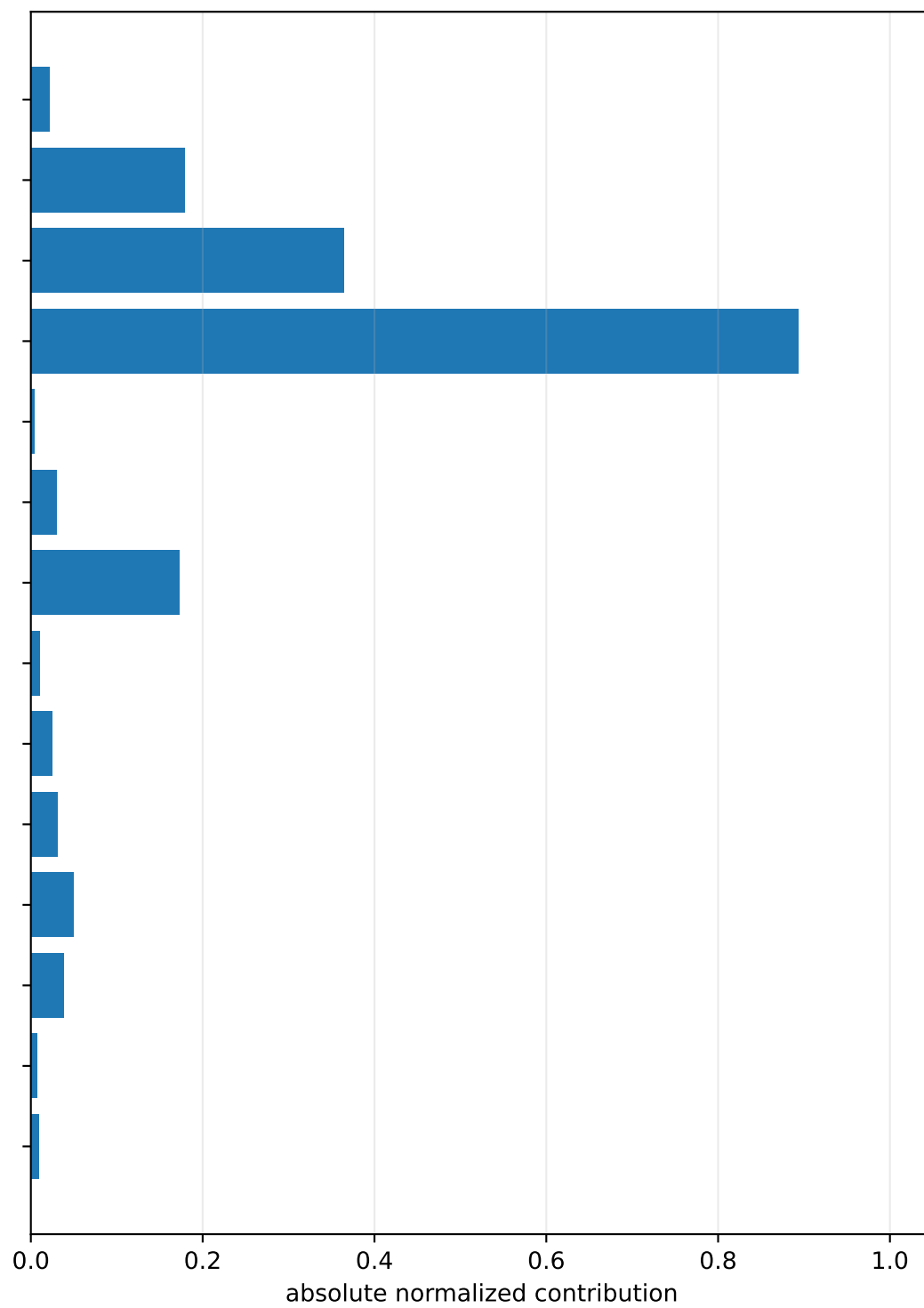


Best and least identified local parameter combinations

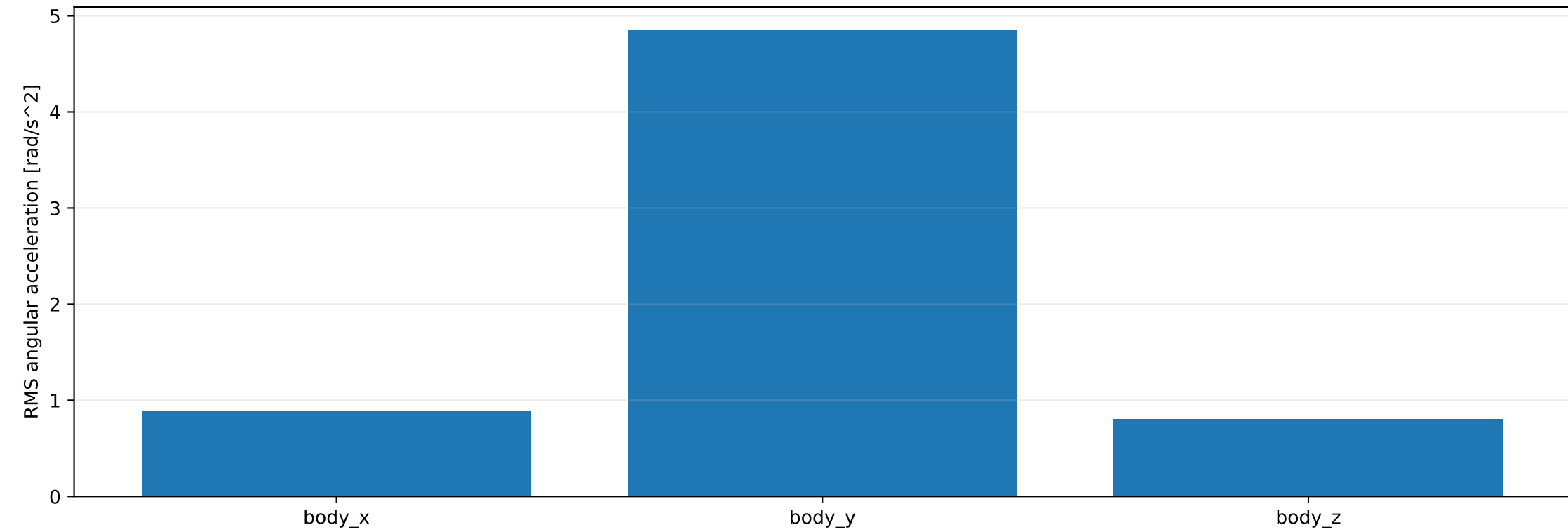
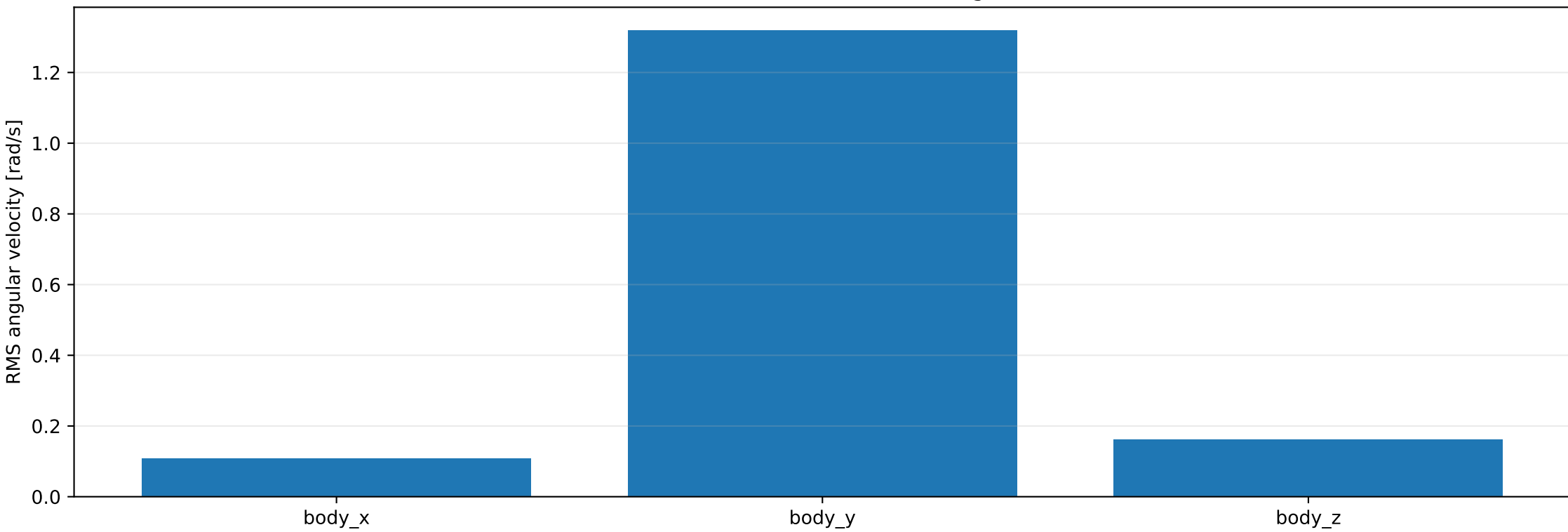
Best identified parameter combination
relative information = 1.000e+00



Least identified parameter combination
relative information = 8.210e-09

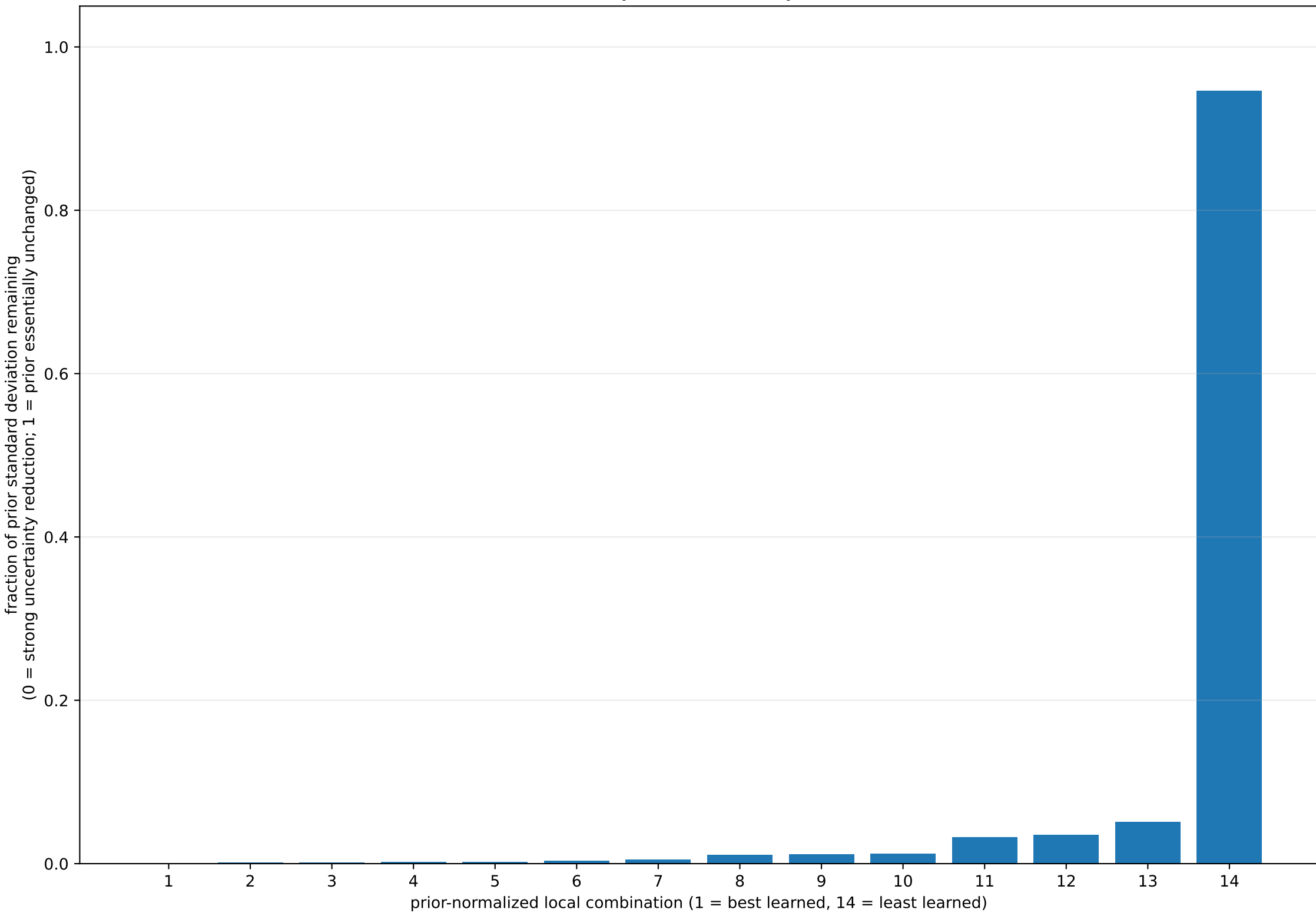


Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

